SHARAN O -241401096 SYSTEM CALLS

Activities Terminal Mar 31 11:32

sharan123@localhost:~/system_calls — /usr/bin/vim pid.c

```
#include <stdio.h>
#include <unistd.h>

int main(){
        printf("PID: %d\n", getpid());
        printf("PPID: %d\n", getppid());

        return 0;
}
```

Activities Terminal Mar 31 11:34

sharan123@localhost:~/system_calls

```
sharan123@localhost ~]$ ls
c.sh  Desktop  Downloads  ifelse.sh  Music  Pictures  Public  REC  sample.sh  scheduling.sh  Templates  while.sh
.sh  Documents  Folder  loops.sh  pc  posneg.sh  qns.txt  sample.c  sample.txt  SFA  system_calls  Videos
sharan123@localhost ~]$ cd system_calls
sharan123@localhost system_calls]$ vi pid.c
sharan123@localhost system_calls]$ gcc pid.c
sharan123@localhost system_calls]$ ./a.out
D:  2802
PID: 2505
sharan123@localhost system_calls]$ vi fork.c
```

Activities Terminal Mar 31 11:35

shjaran123@localhost:~/system_calls — /usr/bin/vim fork.c

```
#include <stdio.h>
#include <unistd.h>

int main(){
        if(fork()==0){
                printf("This is CHILD process\n");
        } else {
                printf("This is PARENT process\n");
        }

        printf("PID: %d\n", getpid());

        return 0;
```

Activities Terminal Mar 31 11:37

sharan123@localhost:~/system_calls

```
[sharan123@localhost system_calls]$ ls
abc.sh  Desktop  Downloads  ifelse.sh  Music  Pictures  Public  REC  sample.sh  scheduling.sh  Templates  while.sh
ab.sh  Documents  Folder  loops.sh  pc  posneg.sh  qns.txt  sample.c  sample.txt  SFA  system_calls  Videos
[sharan123@localhost system_calls]$ vi fork.c
[sharan123@localhost system_calls]$ gcc fork.c
[sharan123@localhost system_calls]$ ./a.out
This is PARENT process
PID: 2824
This is CHILD process
PID: 2825
[sharan123@localhost system_calls]$ vi exec.c
```

Activities Terminal Mar 31 11:39

sharan123@localhost:~/system_calls — /usr/bin/vim exec.c

```
#include <stdio.h>
#include <unistd.h>

int main(){
        printf("Before exec()\n");
        execl("/bin/ls", "ls", NULL);
        printf("After exec()\n");

        return 0;
```

Activities Terminal Mar 31 11:41

sharan123@localhost:~/system_calls — /usr/bin/vim opendir.c

```
#include <stdio.h>
#include <dirent.h>

int main(){
        DIR *dir;

        dir = opendir(".");

        if(dir == NULL){
            printf("Failed to open directory\n");
            return 1;
        }

        printf("Directory opened successfully!!\n");
        closedir(dir);
        return 0;
}
~
```

Activities Terminal Mar 31 11:41

sharan123@localhost:~/system_calls

```
[sharan123@localhost system_calls]$ ls
abc.sh  Desktop  Downloads  ifelse.sh  Music  Pictures  Public  REC  sample.sh  scheduling.sh  Templates  while.sh
ab.sh  Documents  Folder  loops.sh  pc  posneg.sh  qns.txt  sample.c  sample.txt  SFA  system_calls  Videos
[sharan123@localhost system_calls]$ vi exec.c
[sharan123@localhost system_calls]$ gcc exec.c
[sharan123@localhost system_calls]$ ./a.out
abc.sh  exec.c  fork.c  opendir.c  pid.c  readdir.c
ab.sh  exec.c  fork.c  opendir.c  pid.c  readdir.c
[sharan123@localhost system_calls]$ vi opendir.c
```

```
Activities  Terminal                    Mar 31  11:42
sharan123@localhost:~/system_calls
[sharan123@localhost system_calls]$ ls
abc.sh  Desktop   Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh  scheduling.sh   Templates  while.sh
ab.sh   Documents Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  SFA    system_calls Videos
[sharan123@localhost system_calls]$ vi opendir.c
[sharan123@localhost system_calls]$ gcc opendir.c
[sharan123@localhost system_calls]$ ./a.out
Directory opened successfully!
[sharan123@localhost system_calls]$ vi readdir.c
```

```
Activities  Terminal                    Mar 31  11:31
sharan123@localhost:~/system_calls
[sharan123@localhost ~]$ ls
abc.sh  Desktop   Downloads  ifelse.sh  Music  Pictures   Public   REC       sample.sh  scheduling.sh   Templates  while.sh
ab.sh   Documents Folder     loops.sh   pc     posneg.sh  qns.txt  sample.c  sample.txt  SFA    system_calls Videos
[sharan123@localhost ~]$ cd system_calls
[sharan123@localhost system_calls]$ vi pid.c
```